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SECTION: B

QUES12 Develop an inventory management system for different types of products.

1)Product Abstract Class
2) Concrete Product Classes (Electronics, Book, FoodItem)
3) Inventory Class (or Main Method)

Demonstrate adding various product types, updating stock, and calculating total inventory value.

CODE:

```
import java.util.Scanner;
// Abstract Class
abstract class Product {
   protectedStringproductId;
   protected String name;
   protected double price;
   protected int quantityInStock;
   public Product(String productId, String name, double price, int quantityInStock) {
      this.productId = productId;
      this.name = name;
      this.price=price;
      this.quantityInStock = quantityInStock;
   }
   public abstract double calculateValue();
   public void updateStock(int change) {
      quantityInStock += change;
   }
   public void displayProductDetails() {
      System.out.println("ID: " + productId);
      System.out.println("Name: " + name);
      System.out.println("Price: " + price);
      System.out.println("Stock: " + quantityInStock);
   }
}
// Electronics
class Electronics extends Product {
   private int warrantyMonths;
   public Electronics(String productId, String name, double price, int quantityInStock, int
warrantyMonths) {
      super(productId, name, price, quantityInStock);
      this.warrantyMonths = warrantyMonths;
```

```
    }
    public double calculateValue() {
        return price * quantityInStock * (1 + warrantyMonths / 120.0);
    }
    public void displayProductDetails() {
        super.displayProductDetails();
        System.out.println("Warranty: " + warrantyMonths + " months");
        System.out.println("----------------------");
    }
}
// Book
class Book extends Product {
    private String author, isbn;
    public Book(String productId, String name, double price, int quantityInStock, String author,
String isbn) {
        super(productId, name, price, quantityInStock);
        this.author = author;
        this.isbn = isbn;
    }
    public double calculateValue() {
        return price * quantityInStock;
    }
    public void displayProductDetails() {
        super.displayProductDetails();
        System.out.println("Author: " + author);
        System.out.println("ISBN: " + isbn);
        System.out.println("----------------------");
    }
}
// FoodItem
class FoodItem extends Product {
    private String expirationDate;
    public FoodItem(String productId, String name, double price, int quantityInStock, String
expirationDate) {
        super(productId, name, price, quantityInStock);
        this.expirationDate = expirationDate;
    }
    private boolean isExpired() {
        String currentDate = "2026-03-24";
        return expirationDate.compareTo(currentDate) < 0;
    }
    public double calculateValue() {
        return isExpired() ? price * quantityInStock * 0.5 : price * quantityInStock;
    }
    public void displayProductDetails() {
        super.displayProductDetails();
```

```
        System.out.println("Expiry Date: " + expirationDate);
        System.out.println("Expired: " + (isExpired() ? "Yes" : "No"));
        System.out.println("----------------------");
    }
}
// Inventory
class Inventory {
     private Product[] products;
         private int count = 0;
    public Inventory(int size) {
        products = new Product[size];
    }
    public void addProduct(Product p) {
        if (count < products.length) {
            products[count++] = p;
            System.out.println("Product Added!");
        } else {
            System.out.println("Inventory Full!");
        }
    }
    public void removeProduct(String id) {
        for (int i = 0; i < count; i++) {
            if (products[i].productId.equals(id)) {
                for (int j = i; j < count - 1; j++) {
                    products[j] = products[j + 1];
                }
                products[--count] = null;
                System.out.println("Product Removed!");
                return;
            }
        }
        System.out.println("Not Found!");
    }
    public void listAllProducts() { for
        (int i = 0; i < count; i++) {
            products[i].displayProductDetails();
        }
    }
    public double getTotalInventoryValue() {
        double total = 0;
        for (int i = 0; i < count; i++) {
            total += products[i].calculateValue();
        }
        return total;
    }
}
```

```
// Main Class
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Inventory inv = new Inventory(10);
        int choice;
        do {
            System.out.println("\n===== MENU =====");
            System.out.println("1. Add Electronics");
            System.out.println("2. Add Book");
            System.out.println("3. Add Food Item");
            System.out.println("4. Remove Product");
            System.out.println("5. Display All");
            System.out.println("6. Total Value");
            System.out.println("7. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
            switch (choice) {
                case 1:
                    System.out.print("ID: ");
                    String eid = sc.next();
                    System.out.print("Name: ");
                    String ename = sc.next();
                    System.out.print("Price: ");
                    double eprice = sc.nextDouble();
                    System.out.print("Stock: ");
                    int estock = sc.nextInt();
                    System.out.print("Warranty: ");
                    int warranty = sc.nextInt();
                    inv.addProduct(new Electronics(eid, ename, eprice, estock, warranty));
                    break;
                case 2:
                    System.out.print("ID: ");
                    String bid = sc.next();
                    System.out.print("Name: ");
                    String bname = sc.next();
                    System.out.print("Price: ");
                    double bprice = sc.nextDouble();
                    System.out.print("Stock: ");
                    int bstock = sc.nextInt();
                    System.out.print("Author: ");
                    String author = sc.next();
                    System.out.print("ISBN: ");
                    String isbn = sc.next();
                    inv.addProduct(new Book(bid, bname, bprice, bstock, author, isbn));
                    break;
```

```
            case 3:
                System.out.print("ID: ");
                String fid = sc.next();
                System.out.print("Name: ");
                String fname = sc.next();
                System.out.print("Price: ");
                double fprice = sc.nextDouble();
                System.out.print("Stock: ");
                int fstock = sc.nextInt();
                System.out.print("Expiry (YYYY-MM-DD): ");
                String exp = sc.next();
                inv.addProduct(new FoodItem(fid, fname, fprice, fstock, exp));
                break;
            case 4:
                System.out.print("Enter ID to remove: ");
                String rid = sc.next();
                inv.removeProduct(rid);
                break;
            case 5:
                inv.listAllProducts();
                break;
            case 6:
                System.out.println("Total Value = " + inv.getTotalInventoryValue());
                break;
            case 7:
                System.out.println("Exiting...");
                break;
            default:
                System.out.println("Invalid Choice!");
        }
    } while (choice != 7);
    sc.close();
  }
}
```

OUTPUT

```
C:\Users\VIVEK\OneDrive\Desktop>javac Main.java

C:\Users\VIVEK\OneDrive\Desktop>java Main

===== MENU =====
1. Add Electronics
2. Add Book
3. Add Food Item
4. Remove Product
5. Display All
6. Total Value
7. Exit
Enter choice: 1
ID: 1
Name: Laptop
Price:75000
Stock: 20
Warranty: 1
Product Added!

===== MENU =====
1. Add Electronics
2. Add Book
3. Add Food Item
4. Remove Product
5. Display All
6. Total Value
7. Exit
Enter choice: 2
ID: 2
Name: CSE
Price:860
Stock: 30
Author:KireetJoshi
ISBN: Product Added!

===== MENU =====
1. Add Electronics
2. Add Book
3. Add Food Item
4. Remove Product
5. Display All
6. Total Value
7. Exit
Enter choice: 3
ID: 3
Name: Pizza
Price:350
Stock: 20
Expiry (YYYY-MM-DD): 2026-12-27
Product Added!
```

```
===== MENU =====
1. Add Electronics
2. Add Book
3. Add Food Item
4.  Remove  Product
5. Display All
6. Total Value
7. Exit
Enter choice: 4
Enter ID to remove: 3
Product Removed!

===== MENU =====
1. Add Electronics
2. Add Book
3. Add Food Item
4.  Remove  Product
5. Display All
6. Total Value
7. Exit
Enter choice: 5
ID: 1
Name:   Laptop
Price:   75000.0
Stock: 20
Warranty: 1 months
----------------------
ID: 2
 Name: CSE
 Price: 860.0
Stock: 30
Author: Kireet
ISBN: Joshi
----------------------

===== MENU =====
1. Add Electronics
2. Add Book
3. Add Food Item
4.  Remove  Product
5. Display All
6. Total Value
7. Exit
Enter choice: 6
Total Value = 1538300.0

===== MENU =====
1. Add Electronics
2. Add Book
3. Add Food Item
4.  Remove  Product
5. Display All
6. Total Value
7. Exit
Enter choice: 7
Exiting...
```

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QUES13 Develop an inventory management system for different types of products.

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2) **Concrete Product Classes (Electronics, Book, FoodItem)**
3) **Inventory Class (or Main Method)**

Demonstrate adding various product types, updating stock, and calculating total inventory value.

CODE:

```
importjava.util.Scanner;
abstract class LibraryItem {
   protected String itemId, title;
   protected int publicationYear;
   protected boolean isAvailable;
   public LibraryItem(String itemId, String title, int year) {
      this.itemId = itemId;
      this.title = title;
      this.publicationYear = year;
      this.isAvailable = true;
   }
   public abstract void displayItemDetails();
   public boolean borrowItem() {
      if (isAvailable) {
         isAvailable = false;
         return true;
      }
      return false;
   }
   public boolean returnItem() {
      if (!isAvailable) {
         isAvailable = true;
         return true;
      }
      return false;
   }
}
 class Book extends LibraryItem {
   protected String author, isbn;
   public Book(String id, String title, int year, String author, String isbn) {
      super(id, title, year);
      this.author = author;
```

```
        this.isbn = isbn;
    }
    public void displayItemDetails() {
        System.out.println("Book: " + title + " | Author: " + author + "
                | ISBN: " + isbn + " | Available: " + isAvailable);
    }
}
class EBook extends Book {
    private String fileFormat, downloadLink;
    public EBook(String id, String title, int year, String author, String isbn,
                String format, String link) {
        super(id, title, year, author, isbn);
        this.fileFormat = format;
        this.downloadLink = link;
    }
    public boolean borrowItem() {
        System.out.println("Download link: " + downloadLink);
        return true;
    }
    public void displayItemDetails() {
        System.out.println("EBook: " + title + " | Format: " + fileFormat +
                " | Author: " + author + " | Available: " + isAvailable);
    }
}
class AudioBook extends LibraryItem {
    private String narrator;
    private int duration;
    public AudioBook(String id, String title, int year, String narrator, int duration) {
        super(id, title, year);
        this.narrator = narrator;
        this.duration = duration;
    }
    public void displayItemDetails() {
        System.out.println("AudioBook: " + title + " | Narrator: " + narrator + "
                | Duration: " + duration + " mins | Available: " + isAvailable);
    }
}
class DVD extends LibraryItem {
    private String director;
    private int runtime;
    public DVD(String id, String title, int year, String director, int runtime) {
        super(id, title, year);
        this.director  =  director;
        this.runtime = runtime;
    }
    public void displayItemDetails() {
```

```
        System.out.println("DVD: " + title + " | Director: " + director +
                " | Runtime: " + runtime + " mins | Available: " + isAvailable);
    }
}
class Library {
    private LibraryItem[] items = new LibraryItem[20];
    private int count = 0;
    public void addItem(LibraryItem item) {
        if (count < items.length) {
            items[count++] = item;
            System.out.println("Item Added!");
        } else {
            System.out.println("Library Full!");
        }
    }
    public LibraryItem findItemById(String id) {
        for (int i = 0; i < count; i++) {
            if (items[i].itemId.equals(id)) {
                return items[i];
            }
        }
        return null;
    }
    public void borrow(String id) {
        LibraryItem item = findItemById(id);
        if (item != null && item.borrowItem()) {
            System.out.println("Borrowed Successfully!");
        } else {
            System.out.println("Item Not Available!");
        }
    }
    public void returnItem(String id) {
        LibraryItem item = findItemById(id);
        if (item != null && item.returnItem()) {
            System.out.println("Returned Successfully!");
        } else {
            System.out.println("Return Failed!");
        }
    }
    public void listAllItems() {
        for (int i = 0; i < count; i++) {
            items[i].displayItemDetails();
        }
    }
}
public class Main {
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    Library lib = new Library();

    int choice;
    do {
        System.out.println("\n===== MENU =====");
        System.out.println("1. Add Book");
        System.out.println("2. Add EBook");
        System.out.println("3. Add AudioBook");
        System.out.println("4. Add DVD");
        System.out.println("5. Borrow Item");
        System.out.println("6. Return Item");
        System.out.println("7. Display All");
        System.out.println("8. Exit");
        System.out.print("Enter choice: ");
        choice = sc.nextInt();
        switch (choice) {
            case 1:
                System.out.print("ID: ");
                String bid = sc.next();
                sc.nextLine();
                System.out.print("Title:      ");
                String btitle = sc.nextLine();
                System.out.print("Year: "); int
                byear = sc.nextInt();
                sc.nextLine();
                 System.out.print("Author: ");
                 String author = sc.nextLine();
                System.out.print("ISBN: ");
                String isbn = sc.next();
                lib.addItem(new Book(bid, btitle, byear, author, isbn));
                break;
            case 2:
                System.out.print("ID: ");
                String eid = sc.next();
                sc.nextLine();
                System.out.print("Title:      ");
                String etitle = sc.nextLine();
                System.out.print("Year: "); int
                eyear = sc.nextInt();
                sc.nextLine();
                System.out.print("Author: ");
                String eauthor = sc.nextLine();
                System.out.print("ISBN: ");
                String eisbn = sc.next();
```

```
    System.out.print("Format: ");
    String format = sc.next();
    System.out.print("Download Link: ");
    String link = sc.next();
    lib.addItem(new EBook(eid, etitle, eyear, eauthor, eisbn, format, link));
    break;
case 3:
    System.out.print("ID: ");
    String aid = sc.next();
    sc.nextLine();
    System.out.print("Title:      ");
    String atitle = sc.nextLine();
    System.out.print("Year: "); int
    ayear = sc.nextInt();
    sc.nextLine();
    System.out.print("Narrator:    ");
    String narrator = sc.nextLine();
    System.out.print("Duration:   ");
    int duration = sc.nextInt();
    lib.addItem(new AudioBook(aid, atitle, ayear, narrator, duration));
    break;
case 4:
    System.out.print("ID: ");
    String did = sc.next();
    sc.nextLine();
    System.out.print("Title:      ");
    String dtitle = sc.nextLine();
    System.out.print("Year: "); int
    dyear = sc.nextInt();
    sc.nextLine();
     System.out.print("Director: ");
     String director = sc.nextLine();
     System.out.print("Runtime: ");
    int runtime = sc.nextInt();
    lib.addItem(new DVD(did, dtitle, dyear, director, runtime));
    break;
case 5:
    System.out.print("Enter ID: ");
    lib.borrow(sc.next());
    break;
case 6:
    System.out.print("Enter ID: ");
    lib.returnItem(sc.next());
    break;
case 7:
    lib.listAllItems();
```

```
                break;
            case 8:
                System.out.println("Exiting...");
                break;
            default:
                System.out.println("Invalid choice!");
        }
    } while (choice != 8);
    sc.close();
  }
}
```

OUTPUT

```
C:\Users\VIVEK\OneDrive\Desktop>javac Vivek.java
C:\Users\VIVEK\OneDrive\Desktop>java Vivek

===== MENU =====
1. Add Book
2. Add EBook
3. Add AudioBook
4. Add DVD
5. Borrow Item
6. Return Item 7.
DisplayAll
8. Exit
Enter choice: 1
ID: 1
 Title: CSE
 Year: 2006
Author: Ansh
ISBN: 12ds
ItemAdded!

===== MENU =====
1. Add Book
2. Add EBook
3. Add AudioBook
4. Add DVD
5. Borrow Item
6. Return Item 7.
DisplayAll
8. Exit
Enter choice: 2
ID: 2
Title: Porter
Year: 2025
Author: Yash
ISBN: 1dss2
Format: Video
Download Link: sdsnaksa.com
Item Added!
```

```
===== MENU =====
1. Add Book
2. Add EBook
3. Add AudioBook
4. Add DVD
5. Borrow Item
6. Return Item 7.
DisplayAll
8. Exit
Enter choice: 3
ID: 3
 Title: fairy
 Year: 2026
 Narrator: jay
   Duration: 2
Item Added!

===== MENU =====
1. Add Book
2. Add EBook
3. Add AudioBook
4. Add DVD
5. Borrow Item
6. Return Item 7.
DisplayAll
8. Exit
Enter choice: 4
ID: 4
Title: wed
Year: 2027
Director: manu
Runtime: 3
Item Added!

===== MENU =====
1. Add Book
2. Add EBook
3. Add AudioBook
4. Add DVD
5. Borrow Item
6. Return Item 7.
DisplayAll
8. Exit
Enter choice: 5
Enter ID: 3
Borrowed Successfully!
```

```
===== MENU =====
1. Add Book
2. Add EBook
3. Add AudioBook
4. Add DVD
5. Borrow Item
6. Return Item 7.
DisplayAll
8. Exit
Enter choice: 6
Enter ID: 3
Returned Successfully!

===== MENU =====
1. Add Book
2. Add EBook
3. Add AudioBook
4. Add DVD
5. Borrow Item
6. Return Item 7.
DisplayAll
8. Exit
Enter choice: 7
Book: CSE | Author: Ansh | ISBN: 12ds | Available: true
EBook: Porter | Format: Video | Author: Yash | Available: true
AudioBook: fairy | Narrator: jay | Duration: 2 mins | Available: true
DVD: wed | Director: manu | Runtime: 3 mins | Available: true
```